

M359 Hospital domain of discourse summary



This document can be used if you are unsure of the background to the Hospital scenario being developed during the course. It is not referenced explicitly in the course material, but might help clarify any confusion you have regarding the domain of discourse. It should be read in conjunction with the Hospital scenario.

This summary is a record of some of the important details regarding this scenario. If these details had been included in the scenario description they would have obscured the core information being communicated. Some of the details given here would be part of the general knowledge that anyone could be expected to bring to this scenario. Other details are specific to this domain of discourse, and a data analyst would uncover and document this specialist information as part of the analysis process. It is not a complete description – documenting an entire domain of discourse is not possible. In addition, it is expected that a reader might disagree with the ‘reality’ being summarised here. However, during the development of the database, it is important that any disagreements relating to the meaning within the domain of discourse are identified, documented and resolved.

In effect, this summary would be part of the working documents that the analyst would develop during the refinement of the conceptual data model and their exploration of the stakeholders’ requirements. In documenting the analysis leading to the conceptual data model, the design of the database and its implementation, the information in this summary would form part of the data dictionary used throughout the development life cycle.

This summary is in the form of an Entity type catalogue and a Relationship catalogue; sample sets of data that are to be recorded in the database, which are developed to meet the stakeholders’ requirements, are included. You will be taught how to interpret the relationship occurrence diagram notation used in this document in *Block 1* of the course material.

Entity type catalogue

Name: **Ward.**

Description: This is a representation of the properties of a hospital ward in the hospital. Each occurrence represents a different ward.

Attribute details: A ward has a unique name and an identifying ward number. The hospital allocates ward numbers in the range **w1** to **w9** (consequently, limiting the hospital to a maximum of nine wards). Ward names are character strings. Currently, they appear to use regions of the British Isles but this may change. Every ward has at least one bed and there are no wards that have more than 50 beds.

Sample data: **Ward.**

WardNo	WardName	NumberofBeds
w2	Wessex	8
w3	Anglia	10
w4	Lakes	8
w5	Dales	6
w7	Pennines	14

Name: **Patient.**

Description: This represents the properties common to those persons who are patients in the hospital. Each occurrence represents a different patient and includes their name.

Attribute details: A patient is issued an identification number when they are admitted to the hospital. The hospital uses patient identifiers in the range **p01** to **p99** (consequently, this hospital is limited to a maximum of 99 patients). The patient's surname is recorded, and it is not used to identify a patient. A patient's gender is recorded as either **F** for female or **M** for male. Their height is recorded in centimetres to the nearest millimetre, and their weight in kilograms, recorded to the nearest 1/10 of a kilogram.

Sample data: **Patient.**

PatientId	PatientName	Gender	Height	Weight
p01	Thornton	F	162.3	71.6
p07	Tennent	M	176.8	70.9
p08	James	M	167.9	70.5
p09	Kay	F	164.7	53.2
p15	Harris	M	180.6	64.3
p31	Rubinstein	F	155.6	70.1
p37	Boswell	F	172.9	52.6
p38	Ming	M	186.3	85.4
p39	Maher	F	161.9	73.0
p68	Monroe	F	165.0	62.6
p71	Harris	M	186.3	76.7
p78	Hunt	M	179.9	74.3
p79	Dixon	F	163.9	56.5
p80	Bell	F	171.3	49.2
p87	Reed	F	160.0	59.1
p88	Boswell	M	168.4	91.4
p89	Jarvis	F	172.9	53.4

Name: **Nurse.**

Description: This represents the properties common to those people who are nurses in the hospital. Each occurrence represents a different nurse and includes the nurse's name.

Attribute details: A nurse is allocated an identifying staff number by the hospital. Each staff number consists of three digits (from **001** to **999**). No nurse can have the same staff number as a doctor. The nurse's surname is recorded, and it is not used to identify a specific nurse.

Sample data: **Nurse.**

<u>StaffNo</u>	<u>NurseName</u>
113	Hutt
126	Sesonske
362	Colman
153	Cooke
192	Wilkie
206	Reay
424	Cooke
523	Descartes
838	Brown
789	Warnock
834	Boswell
823	Wright

Name: **Team.**

Description: This represents the properties of a collection of doctors. Each occurrence represents a different team and includes the telephone number for contacting that team.

Attribute details: A team is identified by a team code in the range **t01** to **t99**. Each team has a dedicated answerphone service accessible via the internal hospital telephone system, which uses 4-digit numbers.

Sample data: **Team.**

<u>TeamCode</u>	<u>TelephoneNumber</u>
t04	4382
t17	8173
t29	3492

Name: **Doctor.**

Description: This represents the properties of doctors. Each occurrence represents a different doctor; it includes their name and position.

Attribute details: A doctor is allocated an identifying staff number by the hospital. Each staff number consists of three digits (from **001** to **999**). No doctor has the same staff number as any of the nurses. The doctor's surname is recorded, and it is not used to identify a specific doctor. Each doctor has a position (or grade) which is limited to that of a consultant, registrar or house doctor. Registrars and house doctors are sometimes known as junior doctors. Junior doctors and consultants have roles and responsibilities in the hospital.

Sample data: **Doctor.**

StaffNo	DoctorName	Position
110	Liversage	Consultant
131	Kalsi	Consultant
156	Hollis	Registrar
174	Gibson	Registrar
178	Paxton	Registrar
200	Tamblin	Consultant
389	Wright	House Officer
462	Bell	House Officer
493	Rampton	House Officer
517	Goss	Registrar
603	Nolan	House Officer

Name: **Specialist.**

Description: This represents the properties of doctors who have specialisms. Each occurrence represents a different specialist and includes their specialism. Specialist is a weak entity type.

Attribute details: A specialist is a consultant doctor who has a named medical specialism. At present the hospital limits the specialisms of its consultants to **Cardiac**, **Paediatric** or **Orthopaedic**.

Sample data: **Specialist**

StaffNo	Specialism
110	Cardiac
131	Orthopaedic
200	Paediatric

Name: **Treatment.**

Description: This contains details of the treatment given by a doctor to a patient. Each occurrence represents a different treatment given by a doctor to a patient, the date the treatment started and the reason for that treatment to be given. **Treatment** is a weak entity type.

Attribute details: A treatment is identified by the doctor and patient involved in the treatment and the date on which the treatment started. (Note that this does not allow records for the same patient to be treated more than once by the same doctor on the same date.) The reason for treatment is recorded by giving the condition that the treatment is intended to address as described by the doctor treating the patient. The hospital does not record the end date for a treatment (which seems like a strange omission, but they probably have their reasons).

Sample data: **Treatment.**

StaffNo	PatientId	StartDate	Reason
131	p07	9/2/2007	marrow infection
462	p08	3/2/2007	high temperature
156	p09	1/1/2007	low temperature
156	p15	5/2/2007	low temperature
389	p31	9/1/2007	pain relief
517	p37	5/12/2006	pain relief
462	p38	27/12/2006	inflamed joints
603	p38	1/1/2007	low temperature
603	p38	1/2/2007	pain relief
200	p39	11/3/2006	septicaemia
603	p39	1/1/2007	high grade fever
110	p68	11/1/2007	vomiting
603	p88	29/12/2006	high grade fever
178	p89	23/12/2006	pain relief
389	p98	5/2/2007	inflamed joints

Name: **Drug.**

Description: This represents the properties of a drug used by the hospital. Each occurrence represents a different drug; it includes the name of the drug, the type of drug and its price.

Attribute details: The hospital uses an internal code to identify each drug. Each drug is given a 3-character identifier with the first character in the range **A** to **Z**, followed by two digits (e.g. **A00** to **Z99**). The drug is recorded as being of a certain type or category of drug. Both the drug name and the drug type are strings. The unit price of the drug is recorded in pounds and pence. The upper limit of the unit price is £999.99.

Sample data: **Drug.**

DrugCode	DrugName	Type	Price
P02	Agonese	Painkiller	0.02
P27	Acheaway	Painkiller	0.06
P51	Capillucanis	Painkiller	0.04
P73	Dolensol	Painkiller	0.10
B23	Scalagamene	Antibiotic	1.05
B48	Oculustriturol	Antibiotic	0.95
B63	Hallexanuran	Antibiotic	1.01
F28	Disagaric	Antifungal	0.42
F29	Cambogia	Antifungal	0.45
S38	Dormito	Sedative	0.18
S90	Soporatex	Sedative	0.21
X01	Placebo	Placebo	0.01

Name: **Prescription.**

Description: This records the issue of a prescription. Each occurrence represents a different prescription form, which gives details of the daily dosage and total amount prescribed.

Attribute details: The hospital issues a unique number to the prescription each time a drug is given for a treatment. The prescription numbers are in the range **P0001** to **P9999**. The prescription records the number of units of the drug prescribed (in the range **1** to **150**) and the daily dosage of that drug (in the range 1 to 20) to be taken.

Sample data: **Prescription.**

<u>PrescriptionNo</u>	<u>Quantity</u>	<u>DailyDosage</u>
P0328	120	4
P0001	56	8
P0101	100	20
P2121	48	3
P2113	48	4
P8123	28	2
P3472	80	4
P4231	10	1
P8734	14	1
P0005	30	2
P0234	112	8
P3465	21	3
P3342	24	3

Relationship catalogue

Name: **ConsistsOf** (conversely, is a member of).

Description: A team consists of doctors.

Reading: Each team consists of one or more doctors.
Each doctor is a member of zero or one team.

Additional information: **ConsistsOf** and **HeadedBy** are exclusive relationships with respect to the doctor.
A consultant cannot be a member of a team. All other doctors must be members of a team.

Sample relationship occurrences:

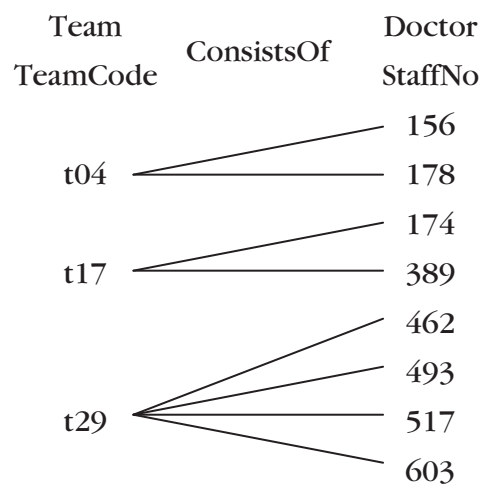


Figure 1

Name: **HeadedBy** (conversely, is head of).

Description: A team has a doctor who is in charge of that team.

Reading: Each team is headed by exactly one doctor.
Each doctor is the head of zero or one team.

Additional information: **ConsistsOf** and **HeadedBy** are exclusive relationships with respect to the doctor.
Each consultant must be the head of a team. Other doctors cannot be the head of a team.

Sample relationship occurrences:

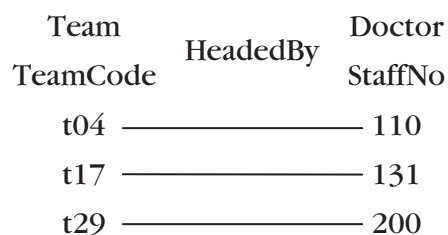


Figure 2

Name: **OccupiedBy** (conversely, is assigned to).

Description: A ward is occupied by patients.

Reading: Each ward is occupied by zero, one or more patients.
Each patient is assigned to exactly one ward.

Sample
relationship
occurrences:

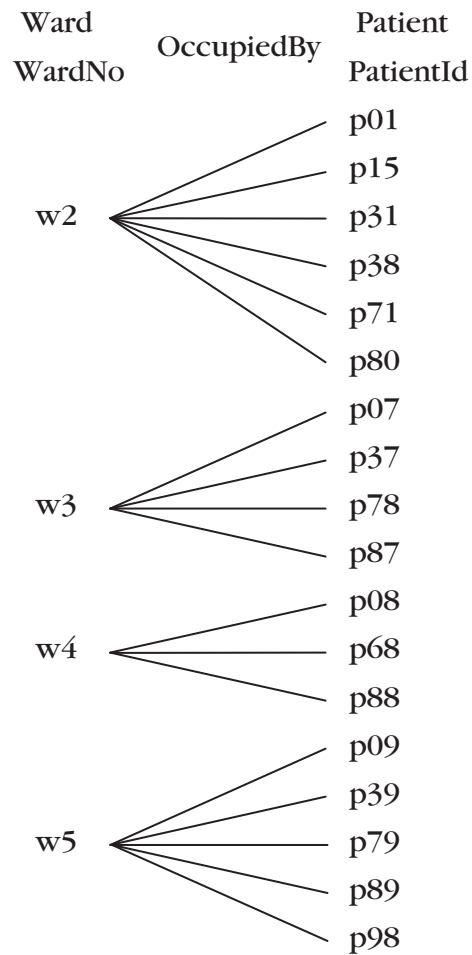


Figure 3

Name: **IsA.**

Description: A specialist is a doctor.

Reading: Each specialist is exactly one doctor.
Each doctor is zero or one specialist.

Additional Information: All consultant doctors are specialists. Other doctors cannot be specialists.

Sample relationship occurrences:

Doctor StaffNo	IsA	Specialist StaffNo
110	————	110
131	————	131
200	————	200

Figure 4

Name: **StaffedBy** (conversely, is allocated to).

Description: A ward is staffed by nurses.

Reading: Each ward is staffed by one or more nurses.
Each nurse staffs exactly one ward.

Sample relationship occurrences:

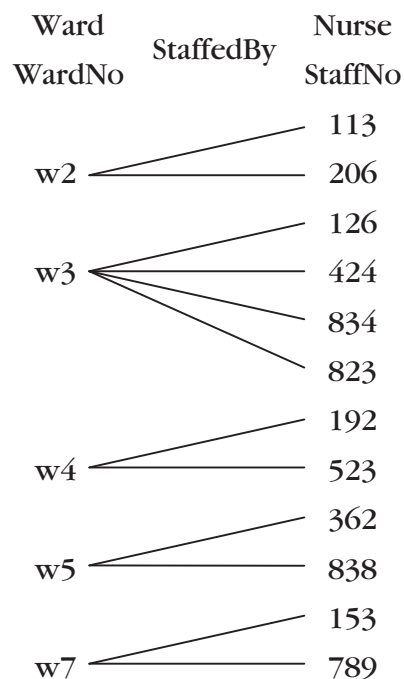


Figure 5

Name: **Supervises** (the relevant roles are supervisor and supervisee).

Description: A nurse can be the supervisor of other nurses.

Reading: Each nurse is the supervisor of zero, one or more nurses.

Each nurse is supervised by zero or one nurse.

Sample
relationship
occurrences:

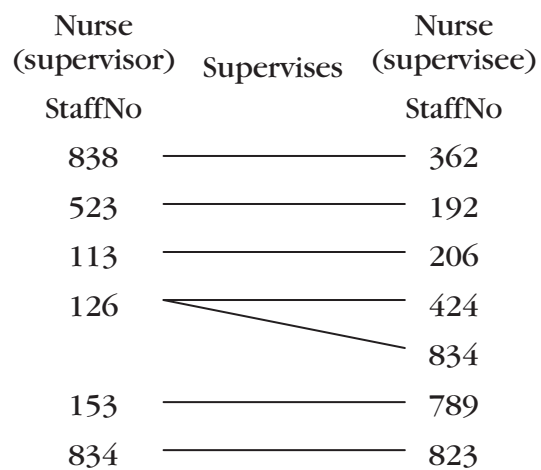


Figure 6

Name: **IsResponsibleFor** (conversely, is under the care of).

Description: A doctor is responsible for the care of patients.

Reading: Each doctor is responsible for zero, one or more patients.
Each patient is under the care of exactly one doctor.

Additional information: Only consultants can be involved in the **IsResponsibleFor** relationship.

Sample relationship occurrences:

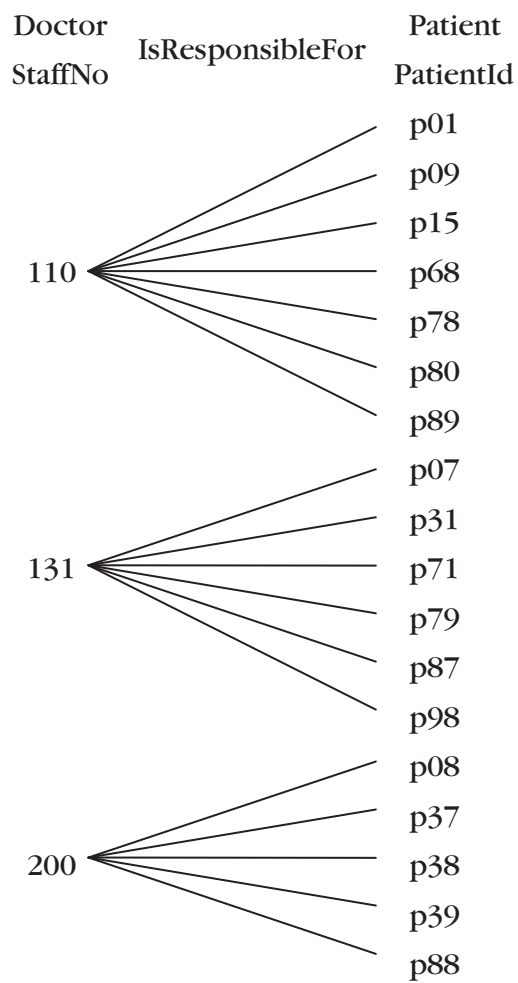


Figure 7

Name: **Provides** (conversely, is provided by).

Description: A doctor provides treatments to patients.

Reading: Each doctor provides zero, one or more treatments.
Each treatment is provided by exactly one doctor.

Sample
relationship
occurrences:

Doctor StaffNo	Provides	StaffNo	Treatment PatientId	StartDate
110	—————	110	p68	11/1/2007
131	—————	131	p07	9/2/2007
156	—————	156	p09	1/1/2007
	—————	156	p15	2/5/2007
178	—————	178	p89	23/12/2006
200	—————	200	p39	11/3/2006
389	—————	389	p31	9/1/2007
	—————	389	p98	5/2/2007
462	—————	462	p08	3/2/2007
	—————	462	p38	27/12/2006
517	—————	517	p37	5/12/2006
	—————	603	p38	1/1/2007
	—————	603	p38	1/2/2007
603	—————	603	p39	1/1/2007
	—————	603	p88	29/12/2006

Figure 8

Name: **Receives** (conversely, is given).

Description: A patient receives treatments from doctors.

Reading: Each patient receives zero, one or more treatments.
Each treatment is given to exactly one patient.

Sample
relationship
occurrences:

Patient			Treatment	
PatientId	Receives	StaffNo	PatientId	StartDate
p07	—————	131	p07	9/2/2007
p08	—————	462	p08	3/2/2007
p09	—————	156	p09	1/1/2007
p15	—————	156	p15	5/2/2007
p31	—————	389	p31	9/1/2007
p37	—————	517	p37	5/12/2006
p38	—————	462	p38	27/12/2006
	—————	603	p38	1/1/2007
	—————	603	p38	1/2/2007
p39	—————	200	p39	11/3/2006
	—————	603	p39	1/1/2007
p68	—————	110	p68	11/1/2007
p88	—————	603	p88	29/12/2006
p89	—————	178	p89	23/12/2006
p98	—————	389	p98	5/2/2007

Figure 9

Name: **Requires** (conversely, is required for).

Description: A treatment requires a prescription which records the drug used for the treatment.

Reading: Each treatment requires zero, one or more prescriptions.
Each prescription is required for exactly one treatment.

Sample
relationship
occurrences:

Treatment			Requires	Prescription	
StaffNo	PatientId	StartDate		PrescriptionNo	
131	p07	9/2/2007	—————	P0328	
178	p89	23/12/2006	—————	P3342	
200	p39	11/3/2006	—————	P8734	
389	p31	9/1/2007	—————	P2121	
462	p38	27/12/2006	—————	P8123	
462	p08	3/2/2007	—————	P0001	
			—————	P0101	
517	p37	5/12/2006	—————	P2113	
603	p38	1/1/2007	—————	P3472	
603	p39	1/1/2007	—————	P0005	
			—————	P0234	
603	p38	1/2/2007	—————	P4231	
603	p88	29/12/2006	—————	P3465	

Figure 10

Name: **IsUsed** (conversely, uses).

Description: A drug is used in a treatment; each use of a drug is recorded on a prescription.

Reading: Each drug is used on zero, one or more prescriptions.
Each prescription uses exactly one drug.

Sample
relationship
occurrences:

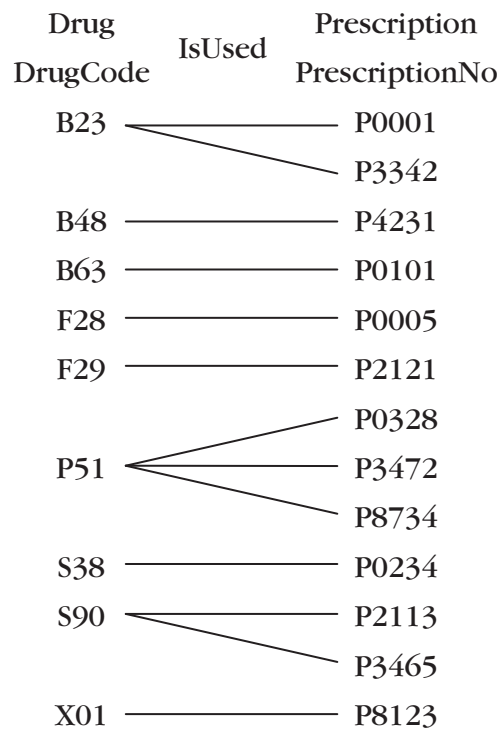


Figure 11